Temperature and Emissivity Retrieval From Hyperspectral Thermal Infrared Data Using Dictionary-Based Sparse Representation for Emissivity

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***Abstract*— The separation of land surface temperature (LST) and land surface emissivity (LSE) is an ill-posed problem in thermal infrared (TIR) remote sensing. By building a new observation matrix to compress the LSE unknown and a dictionary training method to reconstruct complete LSE spectra, a new dictionary-based sparse representation for emissivity (DSRE) method has been proposed to retrieve LST and LSE from the atmospherically corrected hyperspectral TIR data. The proposed method fully utilizes the sparsity of compressed sensing and the empirical knowledge of the trained emissivity dictionary. The sensitivity analysis shows that the modeling accuracies of the proposed method are 0.215 K and 0.0060 for LST and LSE, respectively. Even with the instrument noise of 0.3 K and the uncertainties in atmospheric transmittance, atmospheric upwelling, and downwelling radiance of 10%, the retrieval accuracies are 0.811 K for LST and 0.0241 for LSE, respectively. Then a field experiment was conducted to validate the proposed method, and a comparison was executed to three published methods, including Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) temperature-emissivity separation (TES) (ASTERTES), linear spectral emissivity constraint TES (LSECTES), and iterative spectrally smooth TES (ISSTES). The accuracies of retrieved LST and spectral LSE are 1.41 K/0.009, 2.57 K/0.071, 1.59 K/0.038, and 2.00 K/0.077 for DSRE, ASTERTES, LSECTES, and ISSTES. In contrast to the three published methods, our proposed method is more accurate and effective than other published methods. Especially in the atmospheric absorption band, the proposed method has a strong anti-noise capability to the residuals of environmental downwelling radiance.**

***Index Terms*— Compressed sensing, dictionary-based sparse representation (DSRE), emissivity spectra, hyperspectral thermal infrared (TIR), land surface temperature (LST).**

I. Introduction

As THE direct driving force of longwave radiation exchange in the surface-atmospheric system, land surface temperature (LST) and land surface emissivity (LSE) play fundamental roles in the global and regional processes of physical, chemical, and biological variation on the Earth [1], [2]. LST drives the exchange of long-wave radiation energy between the earth's surface and the atmosphere, and this parameter has been widely applied in many fields such as the research of global warming change, which pervasively affects human life [3], [4]. The LSE can effectively reflect the surface's chemical composition and geometrical structure [5].

Temperature-emissivity separation (TES) is a key problem in quantitative thermal infrared (TIR) remote sensing. However, the accurate retrieval of LST and LSE from space data remains a great challenge [6]. For one thing, the TES is always an ill-posed problem. Because, with the atmospheric parameters known, there will still be $N + 1$ (emissivities of N channels and one temperature) unknowns for the given observed radiance at N channels. One way to solve this ill-posed problem is to add the equations, and another is to decrease the number of unknowns. On the side, accurate atmospheric correction, which may be challenging, is needed in advance of TES, particularly for a high-humid atmosphere [7]. Additionally, the performance of the current TES algorithm is mediocre for the graybody-like surface, such as water and vegetation [8], [9], etc.

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With the rapid development of remote sensing technology, hyperspectral remote sensing, which can provide more spectral information comparing multispectral data, comes into being. Up to now, with the development of aerospace exploration, many spaceborne and airborne hyperspectral TIR sensors have been designed, such as the Infrared Atmospheric Sounding Interferometer (IASI) on the Meteorological Operational (MetOp) satellite [10], the Atmospheric Infrared Sounder (AIRS) on the NASA's Aqua satellite [11], the cross-track infrared sounder (CrIS) on the Suomi National Polar-orbiting Partnership (Suomi NPP) satellite [12], and the Thermal-infrared Airborne Spectrographic Imager (TASI) [13], etc.

To utilize those hyperspectral TIR data effectively, during the last several years, many methods have been proposed to retrieve LST and LSE from earth observation [14]. For example, the Advanced Spaceborne Thermal Emission and Reflection Radiometer TES (ASTERTES) method, which integrates the normalization emissivity method (NEM) [15], the spectral ratio (SR) method [16], and the min-max difference (MMD) methods to realize LST and LSE retrieval, was first proposed by Gillespie et al. [17]. Then Cui et al. [18] and Sobrino et al. [19] applied the ASTERTES method to the airborne hyperspectral TIR data. Under the assumption that the actual surface emissivity spectra are smoother than the atmospheric absorption spectra, an iterative spectrally smooth TES (ISSTES) was proposed by Borel [20]. Based on Wien's approximation of the Planck function, Hook et al. [21] proposed the alpha-derived emissivity method, this method is used first to obtain the relative emissivity spectrum to estimate the LST and LSE. Tang et al. [22] extended the alpha-derived emissivity method by introducing the correction term of atmospheric downwelling radiation, which was posed by Li and Becker [23]. A correlation-based TES (CBTES) method, developed by Cheng et al. [24], utilizes the correlation between atmospheric downward radiance and surface emissivity to optimize LST. Besides, utilizing the residue of atmospheric downwelling radiance in the calculated surface self-emission as a criterion to estimate LST and a stepwise refining method to determine the emissivity at the location of an atmospheric emission line in a narrow spectral region, Cheng et al. [25] came up with the stepwise refining TES (SRTES) method. By approximating the emissivity spectrum using piecewise linear functions, Wang et al. [26] proposed a linear spectral emissivity constraint TES (LSECTES) method. To reasonably reduce the unknown, Zhang et al. [27] introduced the wavelet transform (WT) in TES to propose a new method, the WTTES, which significantly improves the accuracies of LST and LSE.

For the above method, singular points will occur when the instrument noise exists in the observation system or the surface-leaving radiance contains atmospherically corrected residuals [7], [26], and the occurrence of singular points leads to difficulty in finding an acceptable solution in the TES procedure. Furthermore, the degraded performance of published methods on graybody-like surfaces needs to be improved. To overcome these problems, this article proposes a dictionary-based sparse representation for emissivity (DSRE) TES method, which aims to denoise the atmospheric residuals by the sparsity of compressed sensing and recover high-accurate LSE spectral (especially on the graybody-like surface) by the empirical knowledge of trained emissivity dictionary.

The chapters of this article are organized as follows. Section II is devoted to recalling TIR radiative transfer theory and describing the methodology. Section III shows the accuracy and sensitivity of the proposed method with various errors, including the instrument noise, the uncertainty of atmospheric transmittance, upwelling radiance, and dowelling radiance. In Section IV, the field measurements are used to validate the accuracy of the proposed method. In addition, the cross-validation is shown compared to published methods. Finally, the conclusions are drawn.

II. Methodology

A. TIR Radiation Transfer Equation

To briefly describe the physical process of radiation transfer from the surface to the atmosphere, a recall of the background of the TIR spectral region is presented. On simplifying the atmospheric multiscattering effects and according to Kirchhoff's law, the TIR RTE can be written as follows:

$$L_{\mathrm{TOA}}(\lambda) = \tau(\lambda) L_S(\lambda) + L_{\mathrm{AT}}^{\uparrow}(\lambda) \tag{1}$$

$$L_S(\lambda) = \varepsilon(\lambda) B(\lambda, T_S) + [1 - \varepsilon(\lambda)] L_{\mathrm{AT}}^{\downarrow}(\lambda) \tag{2}$$

where λ denotes wavelength, L_{TOA} is the radiance measured at the top of atmosphere (TOA), τ is the atmospheric transmittance, $L_{\mathrm{AT}}^{\uparrow}$ and $L_{\mathrm{AT}}^{\downarrow}$ are atmospheric upwelling and downwelling radiances, respectively, L_S is the surface-leaving radiance, T_S is the LST, ε is the LSE, and B is the Planck's function, which can be expressed as

$$B(\lambda, T) = \frac{C_1 \lambda^{-5}}{e^{C_2/(\lambda T)} - 1} \tag{3}$$

in which both C_1, C_2 are Planck's constants, and $C_1 = 1.191 \times 10^8$ W·μm^4/m^2/Sr, $C_2 = 1.439 \times 10^4$ μm·K.

B. LST and LSE Retrieval Using DSRE

As mentioned earlier, solving the ill-posed problem in TES can through two ways: introducing additional equations or decreasing the number of unknowns. This article proposes a DSRE method based on the compressed sensing theory. In the DSRE method, the emissivity can be represented by a smaller number of sparse coefficients. Therefore, the number of unknowns in TIR RTE decreases accordingly, and the equation can be solved. The DSRE method includes three key steps: 1) emissivity dictionary training; 2) dictionary-based sparse representation and reconstruction for LSE; and 3) iterative solution for LST and LSE.

1) Theory of Compressed Sensing: In compressed sensing, a high-dimensional signal can be recovered with only a few linear measurements, provided the signal is sparse or nearly sparse. And compressed sensing mainly includes three steps, i.e., compressed measurement, sparse representation, and reconstruction calculation [28], [29]. A compressive measurement is performed on a 1-D sparse signal $\mathbf{x}^{L\times 1}$, i.e.,

$$\mathbf{y}^{K\times 1} = \mathbf{\Phi}^{K\times L} \mathbf{x}^{L\times 1} \tag{4}$$

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where $\mathbf{y}^{K\times1}$ is the compressed value of size $K \times L$, $\mathbf{\Phi}^{K\times L}$ is the measurement matrix of size $K \times L$, and $K < L$.

However, the emissivity signals are often not sparse. If donate $\mathbf{x}^{L\times1}$ as the signal without sparsity, a set of basis $\mathbf{\Psi}^{L\times L}$ is needed to be found in N-dimensional space, and the basis can ensure that the coordinate of $\mathbf{x}^{L\times1}$ under $\mathbf{\Psi}^{L\times L}$ is sparse, i.e.,

$$\mathbf{x}^{L\times1} = \mathbf{\Psi}^{L\times L}\mathbf{s}^{L\times1} \tag{5}$$

where $\mathbf{s}^{L\times1}$ is the coordinate of $\mathbf{x}^{L\times1}$; and $||\mathbf{s}^{L\times1}||_0 \ll L$ (the number of non-zero values in $\mathbf{s}^{L\times1}$ is less than L), in other words, $||\mathbf{x}^{L\times1}||_0 \ll L$ under the basis $\mathbf{\Psi}^{L\times L}$.

Based on (5), the essence of sparse representation is to select a $\mathbf{\Psi}^{L\times L}$ through optimization iterations and make $\mathbf{s}^{L\times1}$ has minimum numbers of non-zero. Next, use the reconstruction calculation to obtain the optimal solution of $||\mathbf{x}^{L\times1}||_0$, i.e.,

$$\hat{\mathbf{x}}^{L\times1} = \arg\min ||\mathbf{x}^{L\times1}||_0, \quad \text{s.t. } \mathbf{y}^{K\times1} = \mathbf{\Theta}^{K\times L}\mathbf{s}^{L\times1} \tag{6}$$

$$\mathbf{\Theta}^{K\times L} = \mathbf{\Phi}^{K\times L}\mathbf{\Psi}^{L\times L} \tag{7}$$

where $\hat{\mathbf{x}}^{L\times1}$ is the optimal solution of $\mathbf{x}^{L\times1}$, $\mathbf{\Theta}^{K\times L}$ is the sensing matrix.

Previous research has established that orthogonal transforms, such as fast Fourier transform (FFT), etc., are usually chosen to be the sparse basis [30]. But when a signal $\mathbf{x}^{L\times1}$ contains random noises or missing information, a traditional orthogonal transform is too fixed to approximate the signal sparsely. In recent years, most research in compress sensing has emphasized that the sparse basis can be replaced by an overcomplete dictionary [31], which allows more flexible linear combinations of atoms to have the ability for richer representations. Thus, the overcomplete emissivity dictionary was chosen as the sparse basis for this article.

2) Emissivity Dictionary Training: The overcomplete emissivity dictionary training needs two steps: sparse coding and dictionary updating [32]. This article selected the orthogonal matching pursuit (OMP) and K singular value decomposition (K-SVD) as the sparse coding and dictionary updating algorithms.

OMP is a greedy algorithm that aims to find a sparse representation of the input signal in the form of a linear combination of dictionary atoms [33], [34], [35]. The K-SVD algorithm, integrating SVD and K-means algorithms, proposed by Rubinstein et al. [36], [37], is an iterative optimization procedure that can modify the atoms to fit the input signal. The following steps presented are the way to train an overcomplete emissivity dictionary.

First, preprocess the emissivity spectra extracted from the spectral library such as ASTER 2.0 [38], i.e.,

$$\hat{\boldsymbol{\varepsilon}}_n^{N\times1} = \tilde{\boldsymbol{\varepsilon}}_n^{N\times1} / \text{mean}\left(\tilde{\boldsymbol{\varepsilon}}_n^{N\times1}\right) \tag{8}$$

$$\mathbf{w}_n^{L\times1} = \begin{bmatrix} \hat{\boldsymbol{\varepsilon}}_n^{N\times1} \\ \sqrt[3]{1/\max\left(\tilde{\boldsymbol{\varepsilon}}_n^{N\times1}\right)} \\ \sqrt[3]{1/\min\left(\tilde{\boldsymbol{\varepsilon}}_n^{N\times1}\right)} \end{bmatrix} \tag{9}$$

$$\mathbf{W}^{L\times J} = \left[\mathbf{w}_n^{L\times1}\right]_{n=1}^{J} \tag{10}$$

where the three equations correspond to the flows of preprocess: calculate the ratio spectra $\hat{\boldsymbol{\varepsilon}}_n^{N\times1}$ based on the original emissivity $\tilde{\boldsymbol{\varepsilon}}_n^{N\times1}$ extracted from the spectral library, i is the serial number of selected emissivity, and N corresponding to the channel number is also the sampling number in TIR spectral range; combine two experiential values $\sqrt[3]{1/\max\left(\tilde{\boldsymbol{\varepsilon}}_n^{N\times1}\right)}$, $\sqrt[3]{1/\min\left(\tilde{\boldsymbol{\varepsilon}}_n^{N\times1}\right)}$ and original emissivity as experiential emissivity spectra $\mathbf{w}_n^{L\times1}$; take the experiential emissivity matrix $\mathbf{W}^{L\times J}$, which consists of column vectors $\mathbf{w}_n^{L\times1}$ for J emissivity examples, as the input of dictionary training, and $L = N+2$. Based on many previous experiments, we think the two experiential values are closer to the ratio spectrum, and the three are easier to form a new spectrum.

The second step is to build a measurement matrix $\mathbf{\Phi}^{K\times L}$ to sever the sparse representation for LSE, i.e.,

$$\mathbf{\Phi}^{K\times L} = \mathbf{E}^{K\times L}\left(\mathbf{Q}^{L\times L} * \left(\mathbf{I}^{L\times1}\mathbf{q}_{\text{norm}}^{1\times L}\right)\right) \tag{11}$$

in which

$$\mathrm{E}^{K\times L} = \left[\ \mathbf{E}^{K\times K}\mathrm{O}^{K\times(L-K)}\ \right] \tag{12}$$

$$\mathbf{Q}^{L\times L} = \left[\mathbf{q}_l^{L\times1}\right]_{l=1}^{L} = \left(\mathbf{I}^{J\times1}\left(\mathbf{x}^{L\times1}\right)^{\mathrm{T}}\right)^{\mathrm{T}}\mathbf{W}_{+}^{J\times L} \tag{13}$$

$$\mathbf{q}_{\text{norm}}^{1\times L} = \left[\frac{1}{\left\|\mathbf{q}_l^{L\times1}\right\|_2}\right]_{l=1}^{L} \tag{14}$$

where $\mathbf{I}^{L\times1}$ and $\mathbf{I}^{J\times1}$ are the identity column vector with a length of L and J, respectively, $\mathbf{x}^{L\times1}$ is the original signal, $\mathbf{W}_{+}^{J\times L}$ is the pseudoinverse matrix of $\mathbf{W}^{L\times J}$, the mathematical symbol "$*$" stands for Hadamard product, and $\mathrm{E}^{K\times L}$ is the matrix merged by an identity matrix and a matrix of zeros. $\mathbf{\Phi}^{K\times L}$ will change as the original signal variation, meanwhile, the measurement matrix has the prior information from the training samples. For these characters, the measurement matrix has the merits of adaptivity to signal.

Third, by means of iterative optimization, an overcomplete emissivity dictionary $\mathbf{D}_{\varepsilon}^{L\times L}$ which could sparsely represent well on the $\mathbf{W}^{L\times J}$ can be obtained, i.e.,

$$\mathbf{D}_{\varepsilon}^{L\times L} = \arg\min_{\mathbf{D},\mathbf{X}} ||\mathbf{W}^{L\times J} - \mathbf{D}^{L\times L}\mathbf{S}^{L\times J}||_2 \quad \text{s.t. } \left\|\mathbf{s}_n^{N\times1}\right\|_0 < S, \quad 1 \le n \le J \tag{15}$$

where $\mathbf{S}^{L\times J} = [\mathbf{s}_n^{L\times1}]_{n=1}^{J}$ is the sparse coefficients matrix which is related with $\mathbf{W}^{L\times J} = [\mathbf{w}_n^{L\times1}]_{n=1}^{J}$, $\mathbf{D}^{L\times L}$ is the dictionary to be updated, and in this article we defined $S = 0.618 \times L$ which is the sparsity (the maximum number of non-zero values in $\mathbf{s}_n^{L\times1}$).

Among the emissivity training iteration, the terminal condition is that the $\mathbf{W}^{L\times J}$ and $\mathbf{D}^{L\times L}\mathbf{X}^{L\times J}$ are very close to each other under the measurement by $\mathbf{\Phi}^{K\times L}$. When the iteration terminal condition is met, the dictionary training is complete. Otherwise, based on the OMP algorithm, $\mathbf{S}^{L\times J}$ is recalculated using an unchanged $\mathbf{D}^{L\times L}$, then use K-SVD to update the atoms in $\mathbf{D}^{L\times L}$.

3) Dictionary-Based Sparse Representation for LSE: After the dictionary training, an initial LSE signal $\boldsymbol{\varepsilon}_0^{N\times1}$ with random noises and missing values can be reconstructed using dictionary-based sparse representation.

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First, based on (9), obtain the experiential spectrum $\mathbf{w}_{\varepsilon}^{L\times1}$ for $\boldsymbol{\varepsilon}_0^{N\times1}$, substitute $\mathbf{x}^{L\times1}=\mathbf{w}_{\varepsilon_0}^{L\times1}$ in (13), and calculate $\mathbf{y}^{K\times1}$ based on $\boldsymbol{\Phi}^{K\times L}$. Next, substitute $\boldsymbol{\Psi}^{L\times L}=\mathbf{D}_{\varepsilon}^{L\times L}$ into (7), and subject to $\boldsymbol{\Theta}^{K\times L}$, utilize OMP to approximate $\mathbf{s}_{\varepsilon}^{L\times1}$ in (6). Finally reconstruct a complete LSE spectrum, i.e., $\boldsymbol{\varepsilon}_0^{N\times1}=\boldsymbol{\Lambda}\mathbf{s}_{\varepsilon}^{L\times1}$, in which $\boldsymbol{\Lambda}$ means a series of operations as follows:

$$\begin{bmatrix}\hat{\boldsymbol{\varepsilon}}^{N\times1}\\ \varepsilon_{\max}\\ \varepsilon_{\min}\end{bmatrix}=\mathbf{w}_{\varepsilon}^{L\times1}=\mathbf{D}_{\varepsilon}^{L\times L}\mathbf{s}_{\varepsilon}^{L\times1} \tag{16}$$

$$\boldsymbol{\varepsilon}^{N\times1}=\frac{1}{2}\left(\frac{1/\varepsilon_{\max}^3}{\max(\hat{\boldsymbol{\varepsilon}}^{N\times1})}+\frac{1/\varepsilon_{\min}^3}{\min(\hat{\boldsymbol{\varepsilon}}^{N\times1})}\right)\hat{\boldsymbol{\varepsilon}}^{N\times1}. \tag{17}$$

4) Iterative Solution for LST and LSE: In this article, except for the specific statement, atmospheric correction is assumed to be accurately performed, and the DSRE method mainly focuses on the surface-leaving radiance.

Suppose the surface radiance and atmospheric downwelling radiance are also N-sampling, and donate both as $\mathbf{L}_S^{N\times1}$ and $\mathbf{L}_{\mathrm{AT}\downarrow}^{N\times1}$, respectively. On knowing the measurements of $\mathbf{L}_S^{N\times1}$ and $\mathbf{L}_{\mathrm{AT}\downarrow}^{N\times1}$, to realize the LST and LSE retrieval, the major step is to reduce the LSE unknowns in (2). Through the dictionary-based sparse representation for LSE, except the LST, the unknowns are non-zeros in $\mathbf{s}_{\varepsilon}^{N\times1}$, i.e.,

$$\begin{aligned}\mathbf{L}_S^{N\times1}&=\boldsymbol{\Lambda}\hat{s}_{\varepsilon}^{L\times1}*B(\boldsymbol{\lambda}^{N\times1},T_S)+\left(1-\boldsymbol{\Lambda}\hat{\mathbf{s}}_{\varepsilon}^{L\times1}\right)*\mathbf{L}_{\mathrm{AT}\downarrow}^{N\times1}\\ \text{s.t.}&\begin{cases}\hat{\mathbf{s}}_{\varepsilon}^{L\times1}=\arg\min\left\|\mathbf{s}_{\varepsilon}^{L\times1}\right\|_0\\ \mathbf{y}_{\varepsilon}^{K\times1}=\boldsymbol{\Phi}^{K\times L}\mathbf{D}_{\varepsilon}^{L\times L}\mathbf{s}_{\varepsilon}^{L\times1}\end{cases}\end{aligned} \tag{18}$$

where $\boldsymbol{\lambda}^{N\times1}$ is the vector for wavelengths. In general, $||\mathbf{s}_{\varepsilon}^{L\times1}||_0\ll L$, the number of unknowns is far less than the equations so that the equation can be solved.

Nevertheless, the analytical solutions cannot be directly obtained by (2). Therefore, in our proposed method, the following steps are adopted to retrieve LST and LSE simultaneously from hyperspectral TIR data which is atmospherically corrected:

1) Define the initial value T_0 of LST, for instance, choose the maximum surface brightness temperature among N bands.
2) Calculate the initial LSE $\boldsymbol{\varepsilon}_0^{N\times1}$ by using T_0.
3) Get the reconstructed LSE $\boldsymbol{\varepsilon}_{\mathrm{r}}^{N\times1}$ based on $\boldsymbol{\varepsilon}_0^{N\times1}$ and $\mathbf{D}_{\varepsilon}^{L\times L}$.
4) Establish the cost function, i.e.,

$$\begin{aligned}f=\Big\|&\boldsymbol{\varepsilon}_{\mathrm{r}}^{N\times1}*B(\boldsymbol{\lambda}^{N\times1},T_0)\\&+\left(1-\boldsymbol{\varepsilon}_{\mathrm{r}}^{N\times1}\right)*\mathbf{L}_{\mathrm{AT}\downarrow}^{N\times1}-\mathbf{L}_S^{N\times1}\Big\|_2^2\end{aligned} \tag{19}$$

and utilize the golden-section search method, a technique for finding an extremum of a function inside a specified interval, to refine the T_0 to get a more accurate LST.
5) If the value of f changes by smaller than $1e-5$, or the iteration times are more than 20, the procedure is stopped. Otherwise, return to step 2).
6) Reserve the refined LST T_0 and the reconstructed LSE spectrum $\boldsymbol{\varepsilon}_{\mathrm{r}}^{N\times1}$ as the final retrieval result.

It is worth mentioning that the maximum and minimum constraint of emissivity can be appropriately added in the iteration to improve the calculation speed. In addition, although the DSRE method is slightly more complex, its immunity to instrument noises and atmospheric correction residuals can also ensure that the algorithm can find the optimal solution quickly. We will discuss this characteristic in the further Section III.

C. Evaluation Indicators of Method's Accuracy

To appropriately evaluate the accuracy of proposed and published methods, the following indicators are used in this article. These indicators include the root mean square error (RMSE) between retrieved and actual temperature difference

$$\mathrm{RMSE}_T=\sqrt{\frac{1}{N_S}\sum_{n=1}^{N_S}\left(T_{R,n}-T_{A,n}\right)^2} \tag{20}$$

the mean error (ME) of retrieved and actual temperature difference

$$\mathrm{ME}_T=\frac{1}{N_S}\sum_{n=1}^{N_S}\left(T_{R,n}-T_{A,n}\right) \tag{21}$$

the RMSE between retrieved and actual emissivity for each channel

$$\mathrm{RMSE}_{\varepsilon,i}=\sqrt{\frac{1}{N_S}\sum_{n=1}^{N_S}\left(\varepsilon_{R,n,i}-\varepsilon_{A,n,i}\right)^2} \tag{22}$$

the RMSE of retrieved and actual emissivity spectra difference

$$\mathrm{RMSE}_{\varepsilon}=\sqrt{\frac{1}{N_SN_C}\sum_{n=1}^{N_S}\sum_{i=1}^{N_C}\left(\varepsilon_{R,n,i}-\varepsilon_{A,n,i}\right)^2} \tag{23}$$

where n is the serial number of each sample, i is the serial number of each channel, N_S is the number of total samples, N_C is the number of channels, $T_{R,n}$ and $T_{A,n}$ are the retrieved and actual LSTs, and $\varepsilon_{R,n,i}$ and $\varepsilon_{A,n,i}$ are the retrieved and actual LSEs.

III. Sensitivity Analysis

The method's performance can be evaluated by two aspects: the modeling accuracy of the retrieval method; and the sensitivity to various sources of errors using the simulation data. These errors include the system noises of the on-orbit instrument and the uncertainties of three atmospheric parameters (i.e., atmospheric transmittance, upwelling, and downwelling radiance).

A. Data Simulation

In this article, the data simulation relies on moderate resolution atmospheric transmission (MODTRAN), a mainstream atmospheric radiative transfer model by means of powerful function [39]. To cover as many circumstances as possible, the multiple atmospheric profiles, surface temperature, and emissivity samples were used to simulate hyperspectral TIR data. According to the total column precipitable water vapor

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(TCPWV) at equal intervals, 100 atmospheric temperature, moisture, and ozone profiles were selected from the thermodynamic initial guess retrieval (TIGR) dataset, which is a climatological library of representative atmospheric situations selected by statistical methods and contains a multitude of climate conditions in the tropical, mid-latitudes, and polar regions [40]. These selected profiles cover the bottom atmospheric temperature (donate as T_a) range of 235.85–313.16 K and the TCPWV range of 0.06–5.39 g/cm^2. Based on the bottom atmospheric temperature, the surface temperatures are set from $T_a - 5$ to $T_a + 15$ K with a step of 5 K. In the meantime, 50 diverse surface emissivity spectra (containing water/ice/snow, soil, man-made, vegetation, and rock) extracted from ASTER 2.0 spectral library, which is a collection of contributions in a standard format with ancillary data from the Jet Propulsion Laboratory (JPL), Johns Hopkins University (JHU) and the United States Geological Survey (USGS) [38], are considered. These emissivity spectra are also used to construct the emissivity dictionary.

In addition, the running of MODTRAN relays on other essential parameters. For example, the calculation procedure is executed on the radiance mode with multiple scattering, the aerosol mode is set to typical rural extinction, the viewing zenith angle is 0°, the spectral range is from 8.00 μm (1250 cm^{-1}) to 14.01 μm (714 cm^{-1}) with 1 cm^{-1} spectral resolution, the altitudes of surface and sensor are 0.2 and 100 km, respectively, etc.

Once the simulations are completed, there are 25 000 original data with the different situations for LST and LSE retrieval.

B. Modeling Accuracy of the Proposed Method

Modeling accuracy is the most fundamental indicator of the algorithm's performance. In this section, assuming that there are not any instrument system noise and the atmospheric correction performs absolutely, we analyzed the errors of LST and LSE retrieval for the five surface types separately.

Fig. 1 shows the modeling accuracy of LST retrieval. In this case, the MEs of LST for five types of surfaces are all negative values, which indicates that the retrieval LSTs are generally less than the actual LSTs. The absolute values of LST MEs are less than 0.077 K for any surface type, and the total LST ME is −0.030 K. Thus, it can be thought that the difference between retrieval LSTs and actual LSTs is very small. The LST retrieval RMSEs are 0.145, 0.277, 0.032, 0.248, and 0.227 K for water, soil, man-made, vegetation, and rock. Besides, the total RMSE of retrieved and actual LST difference is 0.215 K.

Fig. 2 presents the comparison between the mean of retrieved and actual LSE spectra for five surface types, and the modeling accuracy of LSE retrieval is also given in this figure. The LSE spectra retrieval RMSEs are 0.0076, 0.0065, 0.0024, 0.0074, and 0.0057 for water, soil, man-made, vegetation, and rock. In addition, the LSE RMSEs in different wavelength positions are also diverse for a type of surface. For water, except for the channels which lie in the range from 12.93 to 14 μm, other channels' errors are not exceeding 0.01. The absorption, scattering, and emission effects of water vapor

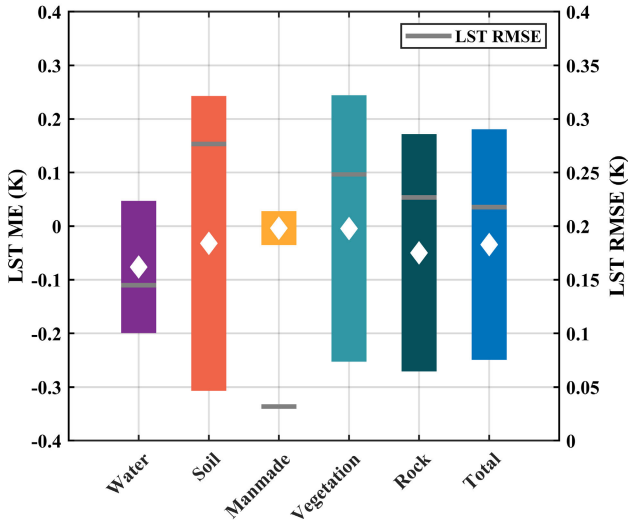


Fig. 1. Modeling accuracy of LST retrieval. The deviation bars are centered on the ME of retrieved and actual LST. The half-length of the bar is equal to the standard deviation (STD) of the difference between retrieved and actual LST. The gray lines are the RMSE of LST, which corresponds to the right Y-axis.

particles probably cause these relatively large errors in the tail of retrieved water LSE spectra. The LSE retrieval RMSEs for the other four types of surfaces are more uniform in wavelength, and the maximum channel RMSEs are 0.0096, 0.0051, 0.0093, and 0.0076 for soil, man-made, vegetation, and rock. And the total LSE RMSE for each channel is not more than 0.0086.

According to the above modeling accuracy, the LST/LSE retrieval RMSE of man-made surfaces is the smallest, but there are no consistent LST errors and LSE errors for the other four types of surfaces.

C. Sensitivity Analysis

A series of experiments were conducted to analyze our proposed method's sensitivity to various errors, including systematic instrument noise and atmospheric parameters bias. Meanwhile, this experiment also can explore the acceptable error limit for the robustness of our method.

1) Sensitivity to the Instrument Noises: First, the method's sensitivity to the on-orbit instrument noises was varied, and the noise with different levels of noise equivalent difference temperature (NEdT) was introduced to the simulated on-orbit spectral radiance. The NEdT levels are 0.0, 0.1, 0.2, and 0.3 K. To make the statistic more reliable, the NEdT was repeatedly added to every retrieval procedure, and the three atmospheric parameters were kept constant. Subsequently, our method was applied to these simulated data.

The LST retrieval histogram (see Fig. 3) shows that the ME of retrieved and actual LST difference has a slight change with the increase of NEdT, and the result of RMSE has a similar statement. Additionally, when the NEdT is less than or equal to 0.2 K, the RMSE of LST retrieval remains constant. Even though the NEdT is equal to 0.3 K, the RMSE is not exceeding 0.28 K. Thus, we can think that the NEdT has a negligible impact on the LST retrieval accuracy.

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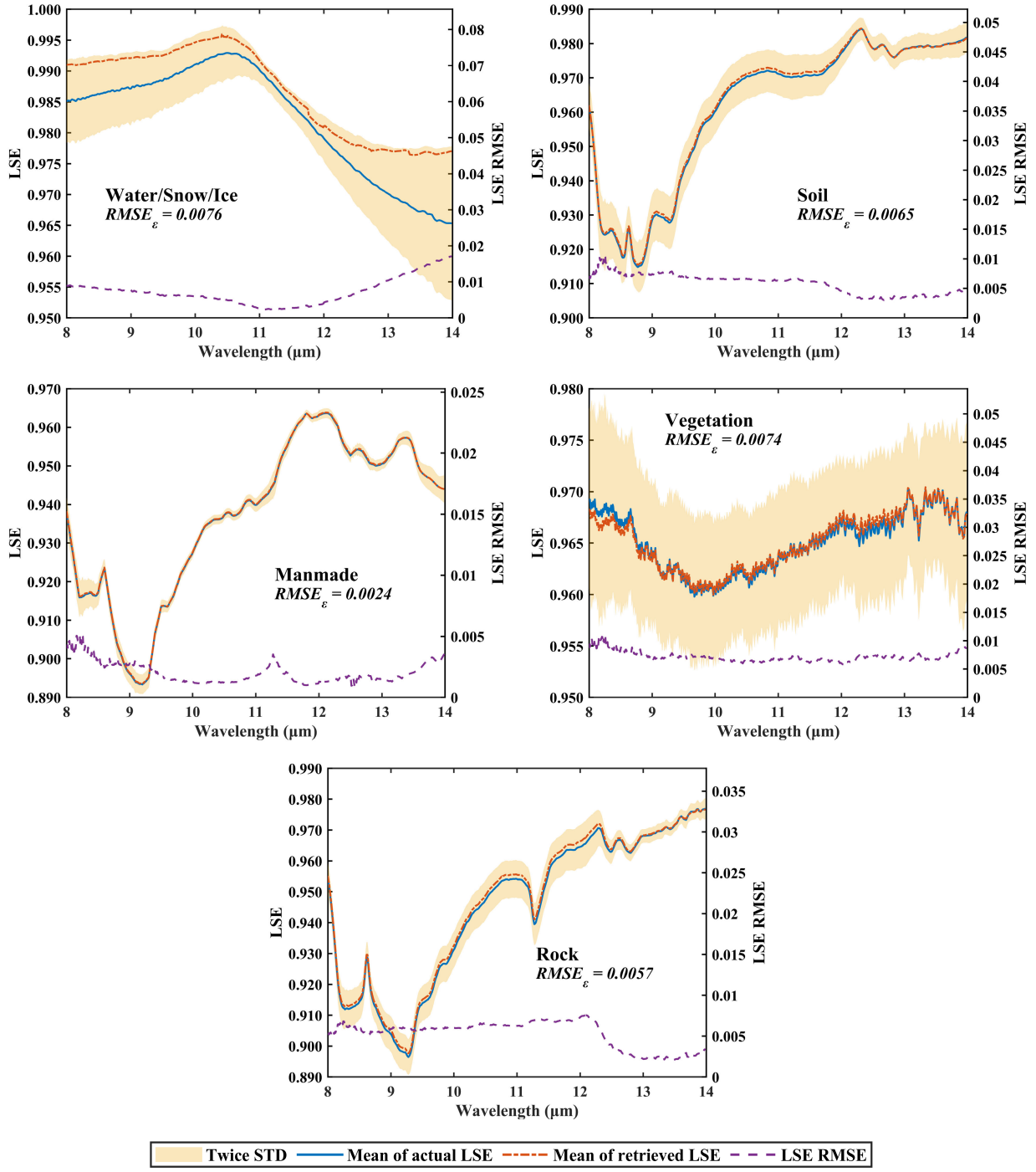


Fig. 2. LSE spectra were retrieved from TOA spectral radiance using our proposed method compared with the laboratory LSE spectra for different surface types. The solid blue lines are the mean of actual LSE spectra from the ASTER 2.0 spectral library. The red chain lines are the mean of the retrieved LSE spectra. The half-width of the yellow band is equal to the STD of the difference between retrieved and actual LSE. The purple dotted lines are the LSE RMSE for each channel, which corresponds to the right Y-axis.

On putting the sight on the spectral LSE retrieval result under different NEdT (see Fig. 4). Like the LST retrieval result, the overall LSE spectra retrieval RMSE are altered slightly as the NEdT increases, i.e., the RMSE of LSE spectra are 0.0060, 0.0063, 0.0070, and 0.0089 under 0.0, 0.1, 0.2, and 0.3 K NEdT. Besides, because of the atmospheric effects, the channels' LSE retrieval RMSEs in the wavelength range 8–8.6 and 12.5–14 μm are visibly more significant than the ones in the most efficient part of the atmospheric window (i.e., 8.6–12.5 μm). There exists a peak of the RMSE graph at 11.3 μm, and we suspect that the height is related to the shape of the LSE because the actual LSE spectra of water, man-made, and rock in the wavelength range around 11.3 μm have a drastic change. When the NEdT = 0.3 K, the maximum

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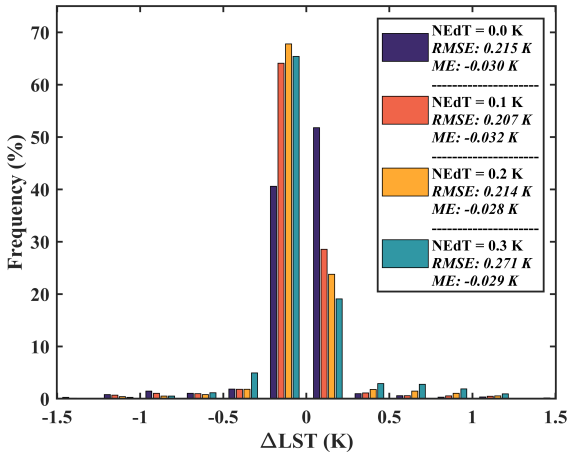


Fig. 3. Histogram of LST errors caused by the instrument noises. The abscissa is the LSE error in Kelvins, and the ordinate is the frequency. The legend shows the RMSEs and MEs of LST for different instrument noises.

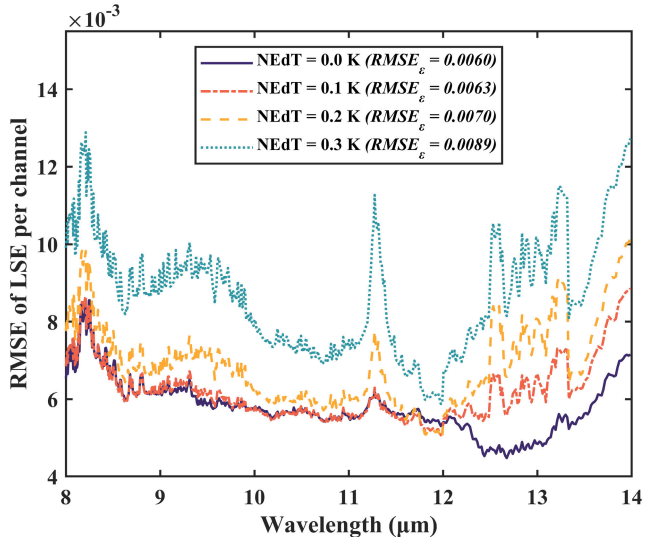


Fig. 4. LSE retrieval RMSE for each channel caused by the instrument noises. The $\mathrm{RMSE}_{\varepsilon}$ is the RMSE of retrieved and actual emissivity spectra difference.

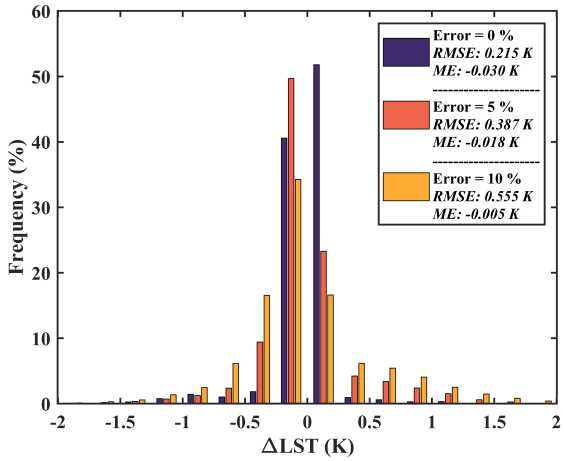


Fig. 5. Histogram of LST errors caused by the uncertainties in atmospheric upwelling radiance. The abscissa is the LSE error in Kelvins, and the ordinate is the frequency. The legend shows the RMSEs and MEs of LST for different instrument noises.

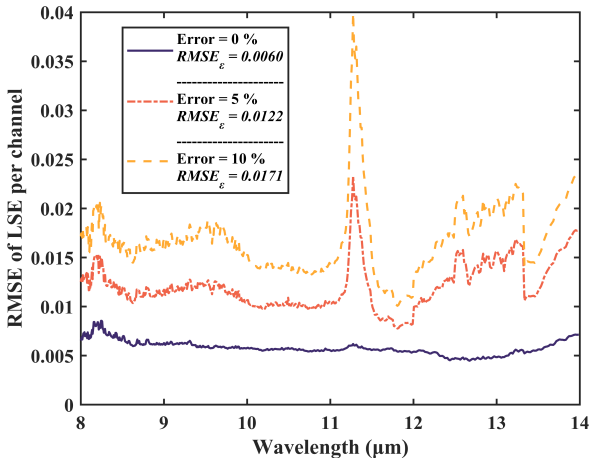


Fig. 6. LSE retrieval RMSE for each channel caused by the uncertainties in atmospheric upwelling radiance. The $\mathrm{RMSE}_{\varepsilon}$ is the RMSE of retrieved and actual emissivity spectra difference.

of channels' LSE retrieval RMSE is not more than 0.0128, apart from the result of 0.3 K NEdT, the LSE accuracy for each channel is within 0.01.

2) Sensitivity to the Uncertainties in Atmospheric Upwelling Radiance: As an unstable and unpredictable system, the atmosphere always influences radiative transfer. Although the precision of atmospheric correction is acceptable, some residual may also exist in the atmospheric parameters. We have explored how atmospheric errors affect the proposed method's accuracy.

In the beginning, the analysis of the method's sensitivity to the uncertainty in atmospheric upwelling radiance was executed. The Gaussian random errors (equal to 0%, 5%, and 10% of the original value) were added to the atmospheric upwelling radiance spectra. Fig. 5 shows the LST retrieval histogram and accuracy under different levels of atmospheric upwelling radiance errors. Apparently, the RMSE of retrieved and actual LST difference gets more significant as the atmospheric error increases. When the error is equal to 10%, the LST retrieval RMSE is 2.58 times as much as the RMSE without any errors, and the histogram distribution also becomes more spread out.

The influence of atmospheric upwelling radiance on the accuracy of LSE retrieval is also huge. Fig. 6 presents the channels' LSE retrieval RMSE caused by the uncertainty in atmospheric upwelling radiance. The spectral LSE retrieval RMSEs are 2.03 and 2.85 times as much as ones without any atmospheric error. And as the adding of error, the channels' LSE RMSE around 11.3 μm becomes more prominent.

3) Sensitivity to the Uncertainties in Atmospheric Transmittance: On adding different levels of errors (i.e., 0%, 5%, and 10% of the original value) into the atmospheric transmittance, we analyzed the sensitivity of our method to the uncertainty in atmospheric transmittance.

Fig. 7 shows the RMSE between retrieved and actual LST under various transmittance errors. Similar to the LST accuracy with the errors in atmospheric upwelling radiance, the errors in transmittance significantly affect the LST retrieval. The LST RMSEs are 0.397 and 0.626 K, corresponding to the

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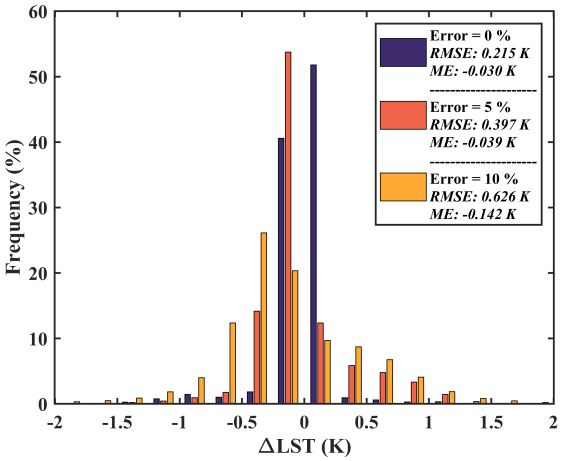


Fig. 7. Histogram of LST errors caused by the uncertainties in atmospheric transmittance. The abscissa is the LSE error in Kelvins, and the ordinate is the frequency. The legend shows the RMSEs and MEs of LST for different instrument noises.

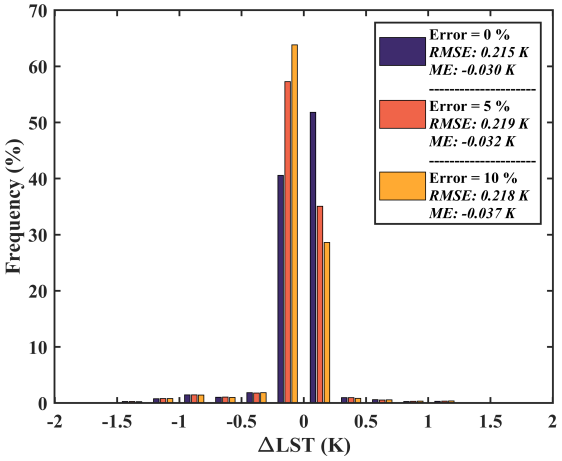


Fig. 9. Histogram of LST errors caused by the uncertainties in atmospheric downwelling radiance. The abscissa is the LSE error in Kelvins, and the ordinate is the frequency. The legend shows the RMSEs and MEs of LST for different instrument noises.

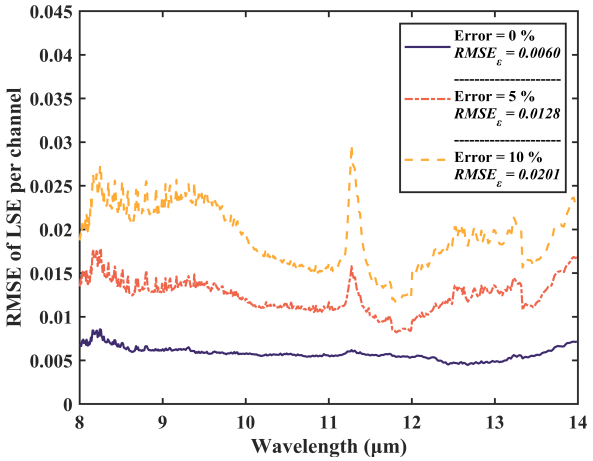


Fig. 8. LSE retrieval RMSE for each channel caused by the uncertainties in atmospheric transmittance. The RMSE_ε is the RMSE of retrieved and actual emissivity spectra difference.

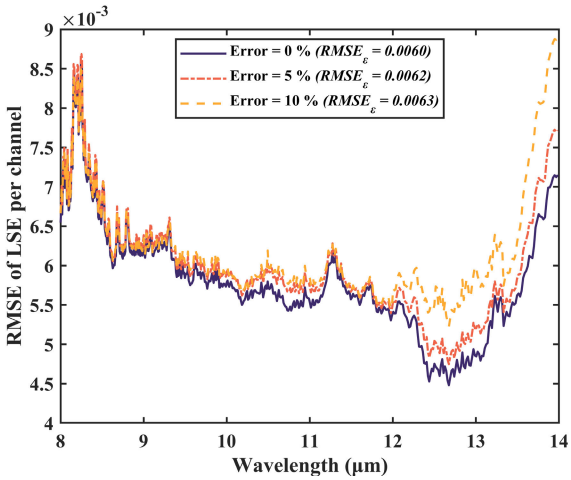


Fig. 10. LSE retrieval RMSE for each channel caused by the uncertainties in atmospheric downwelling radiance. The RMSE_ε is the RMSE of retrieved and actual emissivity spectra difference.

5% and 10% errors. What's more, the impact of errors in atmospheric transmittance on the LSE is equally huge. According to Fig. 8, under 10% Gaussian random error disturbance, most channels' LSE retrieval RMSEs exceed 0.015, and the RMSE between retrieved and actual LSE spectra reaches 0.0201.

4) Sensitivity to the Uncertainties in Atmospheric Downwelling Radiance: In contrast with the analysis results of the method's sensitivity to atmospheric upwelling radiance and transmittance, the proposed method seems not sensitive to the errors in the atmospheric downwelling radiance.

Figs. 9 and 10 present the LST and LSE spectra retrieval accuracy with the effects of different errors, which were randomly added into the atmospheric downwelling radiance spectra. It is evident that although the error is 10%, the LST RMSE has no change, and the RMSE of LSE spectra only increases by 0.0003. We can regard our proposed method as the immunity to the atmospheric downwelling radiance residuals.

5) Method's Accuracy Affected by the Instrument Noises and the Uncertainty in the Three Atmospheric Parameters Simultaneously: Based on the above, it can be concluded that our proposed method's sensitivity to different noises varies. Our method is most sensitive to the errors in the atmospheric transmittance, followed by the atmospheric upwelling radiance residuals and the instrument noises. The method accuracy is least affected by the uncertainty in atmospheric downwelling radiance.

To further explore the simultaneous effects of various errors on our proposed method, another experiment was executed. First, the method's modeling accuracy of LST and LSE retrieval has been known from the previous experiment. Subsequently, when NEdT = 0.1 K, referring to the original value of the simulated atmospheric data, 5% Gaussian random errors were introduced in the spectral atmospheric transmittance, upwelling, and downwelling radiance simultaneously. These data were grouped as dataset A. Then we prepared dataset B, i.e., 10% random errors were added to the original simulated data for three atmospheric parameters, and the NEdT was set

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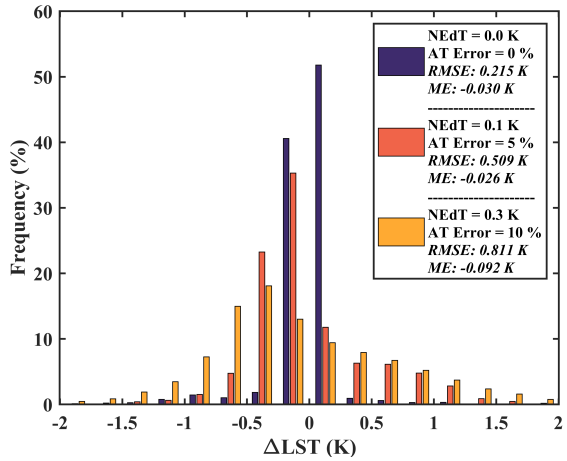


Fig. 11. Histogram of LST errors caused by the instrument noises and the uncertainties in the three atmospheric parameters. The abscissa is the LSE error in Kelvins, and the ordinate is the frequency. The legend shows the RMSEs and MEs of LST for different instrument noises.

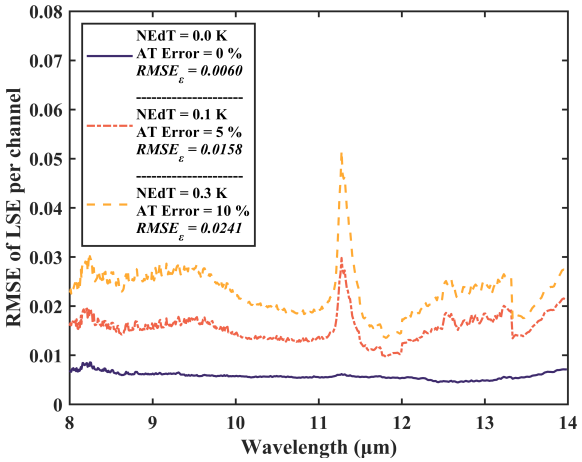


Fig. 12. LSE retrieval RMSE for each channel caused by the instrument noises and the errors of three atmospheric parameters. The RMSE_ε is the RMSE of retrieved and actual emissivity spectra difference.

to 0.3 K. Finally, our TES method was applied to both datasets to retrieve LST and LSE.

What can be seen in Fig. 11 is the rapid growth in the RMSE of LST retrieval as the NEdT and uncertainty in three atmospheric parameters increase. However, although the LST retrieval has been strongly affected by the errors, its accuracy is still higher than 0.82 K. The same statement exists in the LSE retrieval. The graph in Fig. 12 shows that the channels' LSE RMSE are within 0.03, except for the retrieval result lying around 11.3 μm. Moreover, the spectral LSE retrieval RMSE is not larger than 0.025. These results of LST and LSE reveal that our proposed method has relatively strong robustness to the disturbance from instrument noises and atmospheric correction residuals.

IV. Validation With the Field-Measured Data

A. Field-Measured Data

The field-measured data were collected from an Interdisciplinary Research Program on Radiometry in the outdoor environment field experiment site. This field campaign took place at the National Office for Aerospace Studies and Research Center of Fauga–Mauzac, in June 2004, for three weeks [41]. The surface radiance spectra were measured using a BOMEN MR250 series Fourier transform interferometer with a spectral range from 2.86 to 14.29 μm (700–3500 cm^{-1}), 4 cm^{-1} spectral resolution (sampling rate equal to 2 cm^{-1}), and 6 s acquisition time (100 scans averaging). This spectroradiometer, which is equipped AMY08 lens (20 cm diameter) and has a horizontal line of sight, was held at 1.2 m above the ground and pointed to a horizontal surface via a 45° gold mirror (the total path length from the surface to the sensor was about 2 m).

From this experiment's metadata, it can be known that eleven types of surfaces containing natural (water, wood, soil, sand, and stone) and man-made (slate, silicon carbide SiC, and plastic) samples were measured. The surface radiance measurements of ten samples are available to validate our method (see Table I). According to the experimental procedure, A LABSPHERE infragold reflective reference plate was first measured before each observation to reckon the environmental downwelling radiance. A thermocouple was installed inside the reflective panel to record the infragold plate temperature during the experiment. Due to the high infragold reflectance (the average value is more than 0.94 in the TIR range), the surface-leaving radiance of the infragold

TABLE I
Ten Surface Samples Used for Validation [41]

Name	Number	Description
Water	53	Mineral Water
Wood	11	Plywood
Soil	10	Soil from the Negev desert Various grain size < 2 *mm* Composition: SiO_2 (42.2%), CaO (22.8%), Al_2O_3 (5.5%), Fe_2O_3 (2.7%), ...
Slate	10	Homogeneous and flat piece of slate Composition: SiO_2 (60%), Al_2O_3 (17%), Fe_2O_3 (7.6%), K_2O (3.9%), MgO (2.5%), ...
Sand#1	10	Morroco sand. Red color Various grain size < 750 μm Composition: SiO_2 (96.3%), Al_2O_3 (1%), ...
Sand#2	11	Fontainebleau type sand Various grain size < 750 μm Composition: SiO_2 (98.4%), Al_2O_3 (0.6%), ...
Stone#1	10	Flat rough and heterogeneous rock Composition: SiO_2 (97.4%), Al_2O_3 (0.7%), ...
Stone#2	10	Flat rough and homogeneous rock Composition: SiO_2 (77%), Al_2O_3 (12.6%), K_2O (4.6%), Na_2O (3.1%), Fe_2O_3 (1.2%), ...
SiC	11	*SiC* Powder. Grain size ~ 120mm
Plastic	10	Extruded polystyrene

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TABLE II
LST AND LSE SPECTRAL RETRIEVAL RMSE OF OUR PROPOSED METHOD AND PUBLISHED METHODS

Sample Type	Proposed method		ASTERTES		LSECTES		ISSTES	
	LST RMSE (K)	LSE Spectra RMSE	LST RMSE (K)	LSE Spectra RMSE	LST RMSE (K)	LSE Spectra RMSE	LST RMSE (K)	LSE Spectra RMSE
Water	0.5531	0.0064	1.2314	0.0721	0.7959	0.0267	0.5020	0.0473
Wood	1.0609	0.0158	0.7795	0.0500	0.9531	0.0309	1.9833	0.0746
Soil	0.6464	0.0066	0.6076	0.0429	1.0153	0.0377	1.5403	0.0584
Slate	1.7442	0.0026	0.7450	0.0555	1.3780	0.0376	2.7774	0.1015
Sand#1	0.4915	0.0029	0.6029	0.0527	0.8700	0.0325	1.6578	0.0776
Sand#2	0.4047	0.0041	0.4541	0.0554	0.9697	0.0323	1.8264	0.0777
Stone#1	0.5588	0.0130	1.7524	0.0490	1.1847	0.0384	3.9675	0.0767
Stone#2	0.6358	0.0201	1.7735	0.0552	1.1736	0.0386	4.0542	0.1114
SiC	2.9160	0.0055	8.4273	0.1266	3.0344	0.0711	1.8628	0.0510
Plastic	3.4863	0.0036	1.4563	0.0905	3.8476	0.0504	0.7156	0.1413
Total	1.4085	0.0090	2.5733	0.0705	1.5874	0.0379	2.0047	0.0765

plate is very small. Thus, a relatively considerable temperature uncertainty of the infragold plate is tolerance in calculating environmental downwelling radiance. To obtain the actual temperature for validation and monitor the temporal stability of each surface, the surface upwelling brightness temperature was acquired by a HEITRONICS KT19.85, a broadband long-wave infrared (LWIR) radiometer with a spectral response range of 9.6–11.5 μm. Furthermore, an LWIR camera was used to monitor the spatial homogeneity over the spectroradiometer's field of view (FOV). All these three instruments were calibrated on the same M345 MIKRON blackbody, shielded from the wind. More details about this field measurement campaign can be found in [41].

B. LST and LSE Retrieval From Field-Measured Data

This section is devoted to describing the application of the proposed method on field-measured data and the comparison of performance between our method with published methods. In view of the design of the field measurement campaign, the following parameters can be obtained directly, including the BOMEN-measured leaving-surface radiance spectra of the samples and infragold plate, the brightness temperature of the samples' surface measured by KT19.85, the temperature of the infragold plate measured by the thermocouple, and the near actual emissivities of samples and infragold plate measured in the laboratory.

Retrospect the theory of our proposed method, to accurately retrieve LST and LSE from the hyperspectral samples' surface radiance, the environmental (mainly atmospheric) downwelling radiance spectra need to be extracted from the measured radiance of the infragold plate. According to the TIR RTE in (2), the environmental downwelling radiance can be estimated by the following:

$$L_{\mathrm{AT}}^{\downarrow}(\lambda) = \frac{L_G(\lambda) - \varepsilon_G(\lambda) B(\lambda, T_G)}{1 - \varepsilon_G(\lambda)} \tag{24}$$

where L_G, T_G, and ε_G are the surface-emitted radiance measured by the BOMEN spectroradiometer, temperature, and laboratory-measured emissivity of the infragold plate, respectively. Subsequently, the near actual LSTs of samples can be reckoned from the brightness temperatures measured by KT19.85, environmental downwelling radiance, and actual samples' emissivities, i.e.,

$$T_S = B^{-1}\left(\frac{B(T_{\mathrm{KT19}}) - (1 - \varepsilon_{\mathrm{Eff}}) L_{\mathrm{Eff}}^{\downarrow}}{\varepsilon_{\mathrm{Eff}}}\right) \tag{25}$$

where T_{KT19} is the brightness temperature acquired by KT19.85, $L_{\mathrm{Eff}}^{\downarrow}$ and $\varepsilon_{\mathrm{Eff}}$ are the broadband effective environmental downwelling radiance and emissivity, both variables are the results of spectral convolution with the spectral response function (SRF) of KT19.85. It should be noted that two individual look-up tables (LUTs) were used to calculate the actual LSTs. One is the LUT of Planck's function for transforming broadband radiance to temperature, and the other is the LUT between T_S and T_{KT19} in the case where $L_{\mathrm{Eff}}^{\downarrow}$ and $\varepsilon_{\mathrm{Eff}}$ are fixed.

Similar to the simulated data, in this article, we also center around the field-measured data with a spectral range from 8 to 14 μm. In the implementation of validation, for our proposed method, the emissivity dictionary was trained from emissivity spectra extracted from the ASTER 2.0 spectral library and laboratory measurement of field samples. Those emissivity spectra were also totally utilized in constructing the MMD model for the ASTERTES method. Meanwhile, the segment of LSECTES was set to 10 cm^{-1}, and the mean value smoothing method with windows' size of 5 was used in the ISSTES method. Fig. 13 shows an error comparison of LST retrieval between our proposed and published methods with the validating LST for different samples. Table II compares LST and LSE spectra retrieval RMSE of our proposed and published methods.

It can be noted that the total RMSE of retrieved LST is 1.41, 2.57, 1.59, and 2.0 K for our method, ASTERTES, LSECTES, and ISSTES, respectively. There totally are 146 measurements, among which 109 measurements' errors of LST are within 1 K for the proposed method, but only

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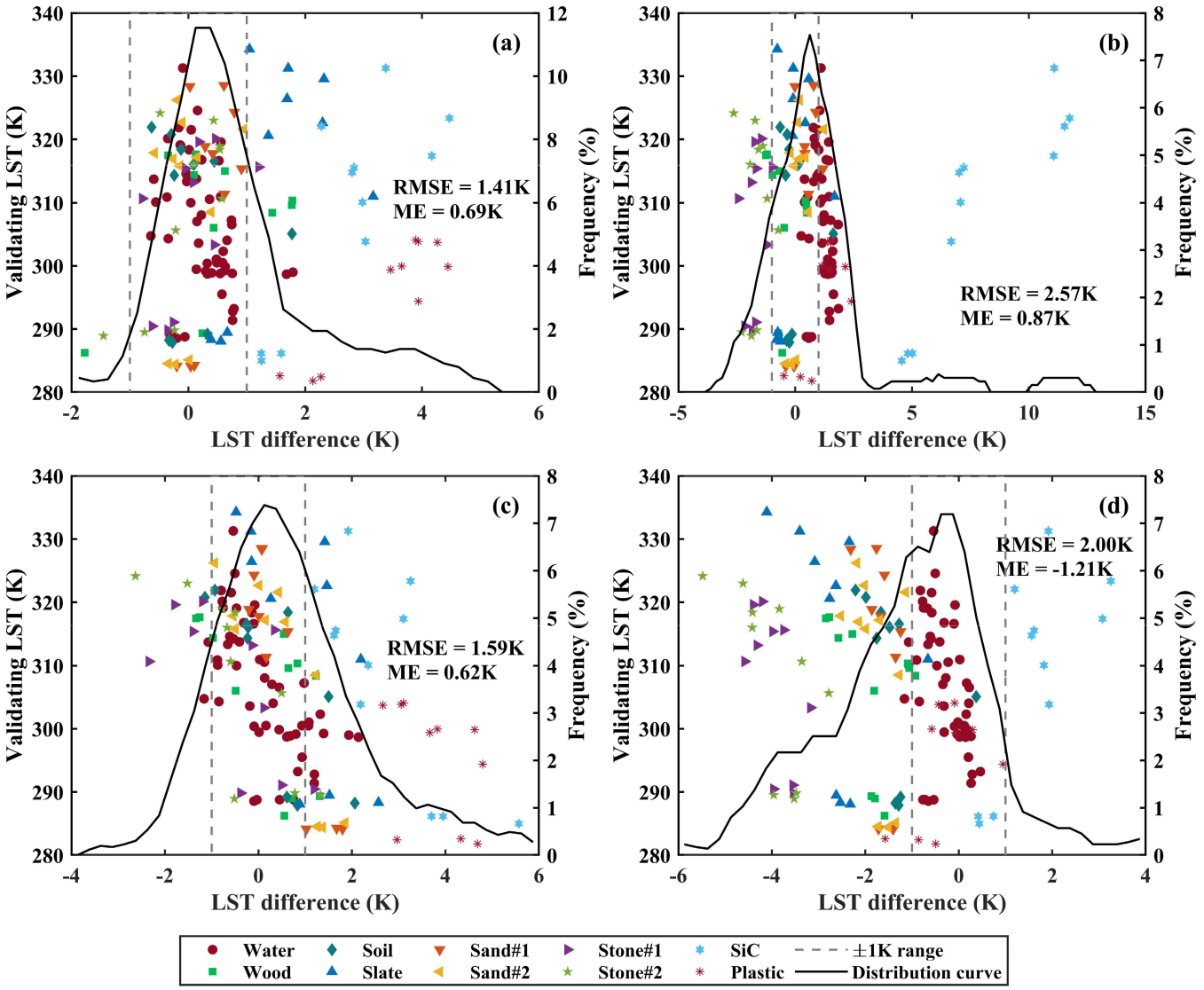


Fig. 13. Difference between retrieval and validating LST for the measurements of ten samples. (a) LST difference for our proposed method, and (b), (c), and (d) for ASTERTES, LSECTES, and ISSTES, respectively.

67, 88, and 65 measurements for ASTERTES, LSECTES, and ISSTES. Obviously, the accuracy of LST retrieval using the DSRE method is more than the other three published methods. Measurements of natural materials are found to have lower LST errors, contrasting with man-made materials. For natural samples, the LST RMSE is 0.62, 1.18, 0.93, and 2.01 K using our method, ASTERTES, LSECTES, and ISSTES; but on applying TES on man-made samples, the LST RMSE is 2.81, 5.11, 2.94, and 1.97 K for each method.

Concentrate on the LST retrieval errors of our proposed method, a few LST errors are more significant than most, which are seemly sample dependent, and some of them exceed 4 K. Only three measurements of the slate sample are observed to have LST errors within 1 K, and entire measurements of SiC and plastic samples are discovered to have LST errors exceeding 1 K. The LST accuracy for each sample can be found in Table II. Excluding these three samples, the RMSE of LST retrieved using the proposed method for the remaining samples is less than 0.62 K.

There are several reasons for the large errors of LST. First, in the field experiment procedure, because of the diversity in FOV between BOMEN and broadband LWIR radiometer, these two instruments do not focus on the same area of samples. Thus, the radiance spectra captured by the sensors of these instruments are different with a high probability. Second, the uncertainty in the emissivity laboratory measurements of the samples can cause the LSTs estimated through the observation of the LWIR radiometer to be inaccurate. This uncertainty may be caused by instrument accuracy, but more comes from the heterogeneity of the sample surface. Naturally, the sample surface tends to be inhomogeneous, and the laboratory emissivity measuring instrument merely has a few square centimeters' views of the sample surface. As a result, the emissivity measured in the laboratory over a few square centimeters cannot represent the whole sample's spectral signature. If the samples are heterogeneous and nonisothermal, the LSTs acquired by these two instruments are different. Third, similar to the laboratory sample emissivities, because

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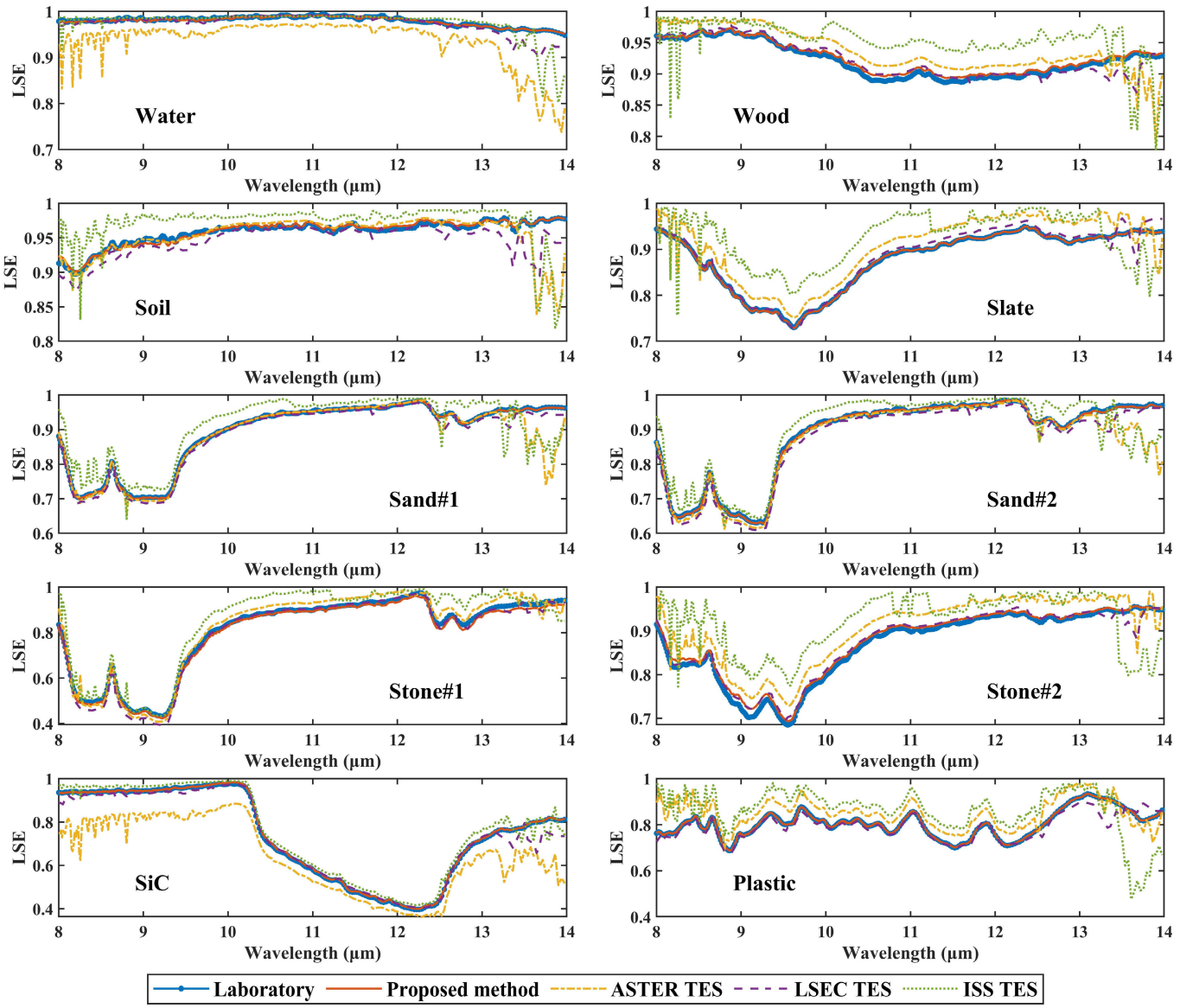


Fig. 14. Comparison between the mean of retrieved LSE and validating LSE. The retrieved LSE spectra were obtained using our proposed method and published methods. And the validating LSE spectra were measured in the laboratory.

the reflective infragold plate's emissivity is also obtained in the laboratory, there probably has uncertainty in the infragold plate's emissivity. In addition, the infragold plate's temperature obtained by the thermocouple may exist bias. These two sources of an error lead to the inaccuracy of environmental downwelling radiance estimation and would indirectly introduce the mistakes into the LSTs estimated by the broadband LWIR radiometer and the spectroradiometer. Undoubtedly, errors in the instrument calibration and LUTs can also lead to a mismatch between the retrieved and validating temperatures.

For each sample, by comparing the laboratory and retrieved emissivity at each channel, we have evaluated the performance of our proposed method and the other three TES methods. In Fig. 14, the laboratory emissivity measurements (the blue line with points) and the mean value of retrieved LSE spectra (other lines with various styles) are plotted in wavelength. Obviously, the LSE spectra retrieved using these four TES methods follow the ones measured in the laboratory to a certain degree. It is also easy to be found that the LSE spectra retrieved by the proposed method are smoother and closer to the validating LSE spectra than the LSE spectra retrieved using the other three TES methods, particularly in the wavelength range (about 8–9 and 12.5–14 μm) where the atmospheric effects have a more significant influence on the LST and LSE retrieval. Based on the accuracy, smoothness, and integrality of retrieved LSE spectra, the performance orders from high to low are the following: the proposed method, LSECTES, ASTERTES, and ISSTES. These orders can also be verified by the LSE retrieval RMSE for each channel (see Fig. 15) and each method's total LSE spectra retrieval RMSE (see Table II). It is worth noting that our method's total RMSE of LSE spectra retrieval is about 0.0090, but the RMSE is 0.0705, 0.0379, and 0.0765 for ASTERTES, LSECTES, and ISSTES methods. On excluding the result with larger errors for wood and stone

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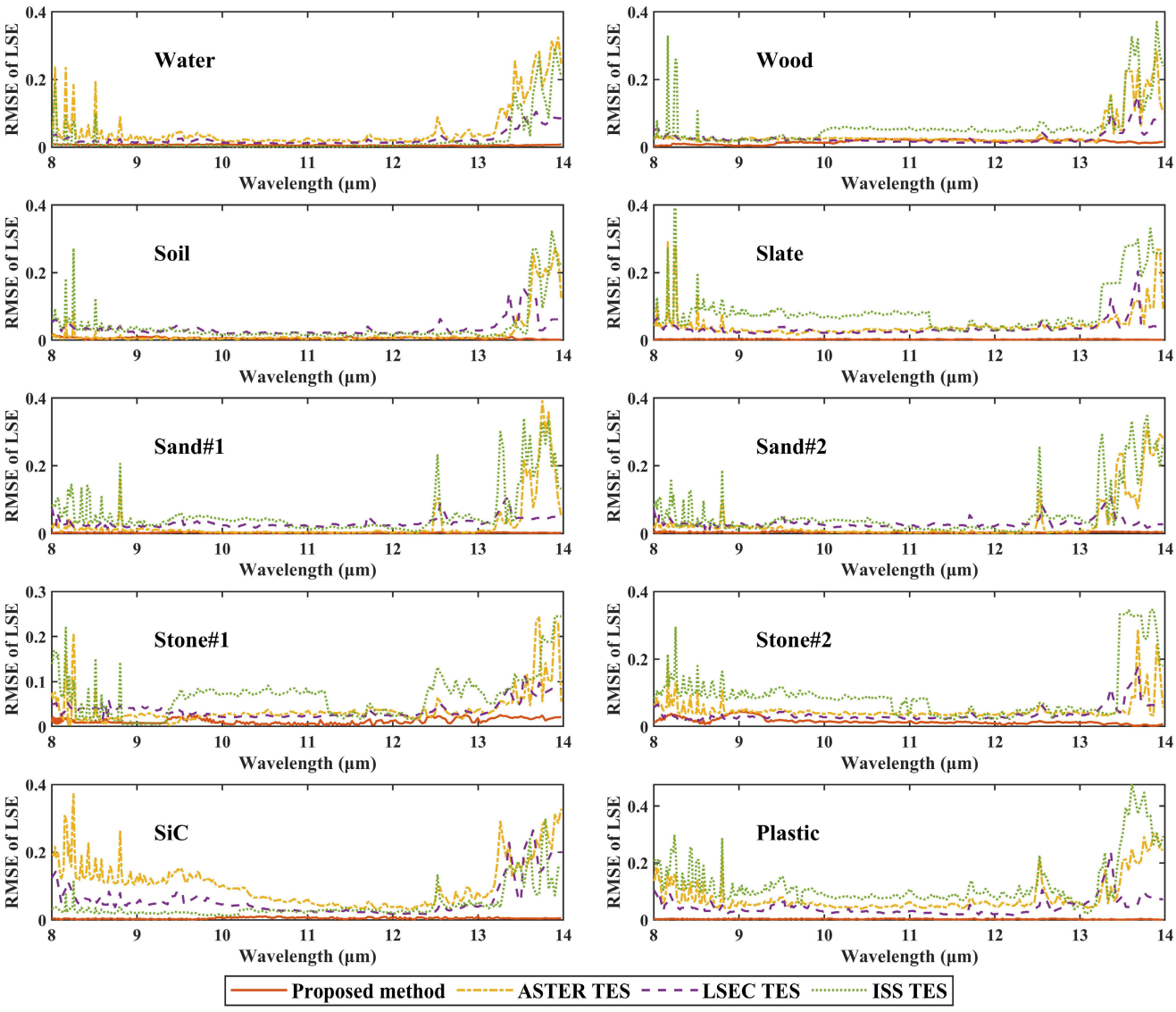


Fig. 15. RMSE for each channel between retrieved LSE spectra and the laboratory LSE difference. The retrieved LSE spectra were obtained using our proposed method and published methods.

samples, the overall RMSE of LSE spectra retrieved by this article's method is 0.0054. Additionally, for water samples, the RMSE of LSE spectra for the DSRE method is no more than 0.0070, but the accuracies for the other three published mothed are worse than 0.0260. This comparison proves that our proposed method also improves the TES performance on the graybody-like surface.

Due to instrument performance and atmospheric effects being diverse at different wavelengths, the retrieved LSE accuracy for each channel is different. On the one hand, the instrument performance mainly relies on the device processing precision and instrument calibration accuracy; on the other hand, the atmospheric effects include emission, absorption, and scattering by various atmospheric particles, for example, water vapor, ozone, and carbon dioxide absorption bands existing at about 8–9, 9–10, and 12.5–14 μm, respectively. Refer to Fig. 15, for each sample, after executing the retrieval procedure using ASTERTES, LSECTES, and ISSTES, it is evident that the LSE RMSE at the range from 10 to 12 μm far less than other wavelength positions; on the contrary, the LSE accuracy of our method is roughly the same at each wavelength position. We think that the reason why the published methods' LSE retrieval accuracies at the beginning and ending TIR bands are worse than other spectral positions (i.e., some larger value occurred in Fig. 15) is probably that the accuracies are affected by environmental downwelling radiance residuals, which could lead to the singular values occurring in the LSE estimation. If ignore the LSE retrieval results with significant errors (lying in the range of 8–8.8 and 13.3–14 μm), the RMSEs of LSE spectra are 0.0091, 0.0403, 0.0277, and 0.0479 for DSRE, ASTERTES, LSECTES, and ISSTES. This contrast again confirms that the proposed method has higher robustness to the instrument's systematic noises and residuals in environmental downwelling radiance.

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V. Conclusion

As vital parameters which can reflect the earth's surface thermal radiation ability, LST and LSE retrieval from aerospace data has been studied for several years. Nowadays, the use of hyperspectral TIR sensors' data is a sufficient route to realize LST and LSE retrieval with high accuracy because they can provide rich spectral dimension information. To utilize the detailed spectral information to eliminate the underdetermined problem in retrieval and get a complete LSE spectrum, a new TES method, i.e., DSRE, has been proposed in this article to retrieve LST and LSE spectra from atmospherically corrected hyperspectral TIR data. This method takes advantage of sparsity and empirical knowledge from dictionary-based sparse representation. Meanwhile, the measurement matrix and the dictionary training procedure came up by us make the proposed method a perfect fit for LSE spectra retrieval.

A series of sensitivity has been carried out assuming that the sensor's performance is not perfect or the atmospheric correction is not complete, and the following conclusions can be made. First, our method has relatively high modeling accuracy from simulated data. In this article, without any error disturbance, the total RMSEs are 0.215 K and 0.0060 for LST and LSE retrieval, and the LSE RMSE for each channel is not more than 0.0086. Second, for LST retrieval, when NEdT = 0.3 K, the RMSE is 0.271 K; 10% Gaussian random errors in atmospheric transmittance, upwelling, and downwelling radiance, respectively, cause an RMSE on the LST retrieval of 0.626, 0.555, and 0.218 K. Besides, for LSE retrieval, the RMSE between retrieved and actual LSE spectra difference is 0.0089 at NEdT = 0.3 K; when 10% uncertainties in atmospheric transmittance, upwelling and downwelling radiance, respectively, the RMSEs of LSE spectra are 0.0201, 0.0171, and 0.0063. In contrast to the sensitivity analysis to the instrument noises and the random errors in atmospheric dowelling radiance, our method is more sensitive to the uncertainty in atmospheric transmittance and upwelling radiance. Third, although under the circumstance with NEdT = 0.3 K and 10% uncertainties in the three atmospheric parameters simultaneously, the retrieval accuracy is not worse than 0.811 K and 0.0241 for LST and LSE. This result indicates that our proposed method has stronger robustness to the instrument noises and the residuals of atmospheric correction. Nevertheless, we still recommend that our method requires good atmospheric correction. If the atmospheric correction does not perform well, the accuracy of LST and LSE spectra maybe deteriorate and become unacceptable. The good news is that integrated inversion is expected to solve this problem. Without any prior knowledge, the simultaneous retrieval of atmospheric and land surface parameters, which is the long-term goal of scientific researchers, has achieved promising results under different land surface and atmospheric conditions [42].

Application of our proposed method to the field measurement shows that a 1.41 K RMSE of LST can be obtained, and the RMSEs of LST are 2.57, 1.59, and 2.0 K for ASTRETES, LSECTES, and ISSTES methods. Moreover, our method's total RMSE of LSE spectra retrieval is less than 0.0090, but the RMSE is 0.0705, 0.0379, and 0.0765 for ASTERTES, LSECTES, and ISSTES methods. Compared with other published methods, our method is more precise and efficient.

Finally, we think that further studies should be carried out. First, the OMP algorithm in compressed sensing is too simple to fit various emissivities signals, so the modification of its optimization algorithm is considered. Second, more field experiments should be executed to validate the proposed method. Finally, as the current hyperspectral TIR satellite is not specially designed for Earth observation and its data have a low spatial resolution, the physical meaning of component LST and LSE on the sub-pixel scale should be defined clearly.

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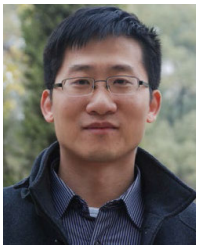
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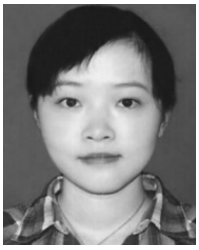
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